



THE STATE UNIVERSITY OF ZANZIBAR

SCHOOL OF COMPUTING, COMMUNICATION AND MEDIA STUDIES

**DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION
TECHNOLOGY**

Software Design Document (SDD)

QueuExpress System

Prepared by:	Hafidh Mwita Haji
Student ID:	BITAM/11/23/099/TZ
Supervisor:	Dr. Abdulrahim Haroun Ali

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INRODUCTION

1.1 Purpose and Scope

The purpose of this System Design Document (SDD) is to provide a detailed design description of the **QueueXpress** System, a digital queue management solution developed to improve service delivery efficiency in Yas telecommunication shop.

This document translates the system requirements defined in the Software Requirements Specification into a structured system design that can be used to implement the system. It covers system architecture, database design, user interfaces, and system interactions.

1.2 Project Executive Summary

QueueXpress is a mobile-based queue management system designed to replace traditional and manual queuing systems with a modern digital solution.

The system allows customers to join queues by scanning a QR code, monitor their queue position in real time, and receive notifications when their turn is approaching. Staff members manage the queue using a dashboard, while administrators configure system settings and monitor performance.

The system is built using React.js for web interface, React native for mobile application, Django for logic and MySQL for database and storage

The design focuses on simplicity, scalability, and efficiency, ensuring that it can operate effectively even in environments with limited infrastructure.

1.2.1 System Overview

QueueXpress follows a three-tier architecture where users interact with the system through web or mobile interfaces. Main Components are

Presentation Layer: This is a customer mobile interface, Staff dashboard and Admin dashboard

Application Layer: This is where the business logic a handled like Manages queue operations and Processes requests

Database Layer: This is where the Stores users, queues, batches, and logs **is** stored

Core Workflow

1. Customer scans QR code at service center entrance using smartphone camera
2. Mobile browser opens QueueXpress and adds customer to queue
3. Customer receives queue number and batch number instantly
4. Customer views real-time status including people ahead and estimated wait time
5. Staff calls next customer from dashboard when ready
6. System notifies customer through visual update on status page
7. Staff serves customer and marks as completed
8. Customer submits optional feedback after service

1.2.2 Design Constraints

The system design is subject to the following constraints:

- Limited infrastructure. display screens for showing queue progress to customer
- Dependence on internet connectivity
- Limited technical skills of end users
- Academic project time constraints

Assumptions Made:

- The service center has reliable internet connectivity
- Customers have smartphones with camera and internet access
- Staff have access to computers or tablets for dashboard
- Daily customer volume does not exceed 500 customers per day
- QR code will remain physically displayed

1.2.3 Future Contingencies

The following potential changes may affect system design:

- Addition of appointment booking feature

- Expansion to support multiple branches
- Upgrade to real-time communication using WebSocket
- Integration with push notification systems (SMS, push notifications)
- Response time analytics and staff performance tracking
- Upgrade to real-time communication using WebSocket

1.3 Document Organization

This document is organized into the following sections:

Section 1: Introduction

Section 2: System Architecture

Section 3: Database Design

Section 4: Human-Machine Interface

Section 5: Detailed Design

Section 6: External Interfaces

Section 7: System Integrity Controls

Each section provides detailed information necessary for system development and implementation.

1.4 Points of Contact

- i. Project Developer for Project Development
- ii. Supervisor for Project guidance and evaluation
- iii. End Users for Provide requirements and feedback

1.5 Project References

Software Requirements Specification (SRS) Document

MySQL Documentation

Django (Version 6.0) [Computer Software]. (2026). Retrieved from <https://www.djangoproject.com/>.

Meta Platforms, Inc. (n.d.). React. Retrieved April 20, 2026, from <https://react.dev/>

Meta Platforms, Inc. (2024). *React Native* (Version 0.73) [Computer software].

<https://reactnative.dev/>

1.6 Glossary

Appendix A: Glossary

Terms	Definition
API	Application Programming Interface
ERD	Entity Relationship Diagram
SRS	Software Requirements Specification
System Setting	Configurable parameters such as batch size and working hours
Dashboard	Interface used by staff/admin to monitor and manage the system
Notification	A system alert sent to customers regarding queue status
Counter	A service point where staff attend customers
Priority Service	Special handling of urgent customers by staff
QueueXpress	The proposed digital queue management system
Customer	A user who joins the queue to receive service
Staff	A service provider responsible for attending customers
Admin	A system administrator responsible for managing and configuring the system
Queue	A sequence of customers waiting for service
Batch	A group of customers allowed to be near the service area at a given time
QR Code	A machine-readable code used for quick access to the queue system
Check-in	The process of confirming arrival at the service center
No-show	A customer who fails to appear when called
WebSocket	Protocol for real-time bidirectional communication
ORM	Object-Relational Mapping (Django's method for database interaction)

SYSTEM ARCHITECTURE

2.1 System Hardware Architecture

The QueueXpress system operates using standard computing devices and does not require specialized hardware infrastructure.

Hardware Components

Client Devices: This is used by customers, staff, and administrators to access the system. This includes Smartphones for customers, Desktop/Laptop computers for staff and admin

QR Code Display: Printed QR code or digital display which is used by customers to join the queue

Network Infrastructure: Internet connection required for communication

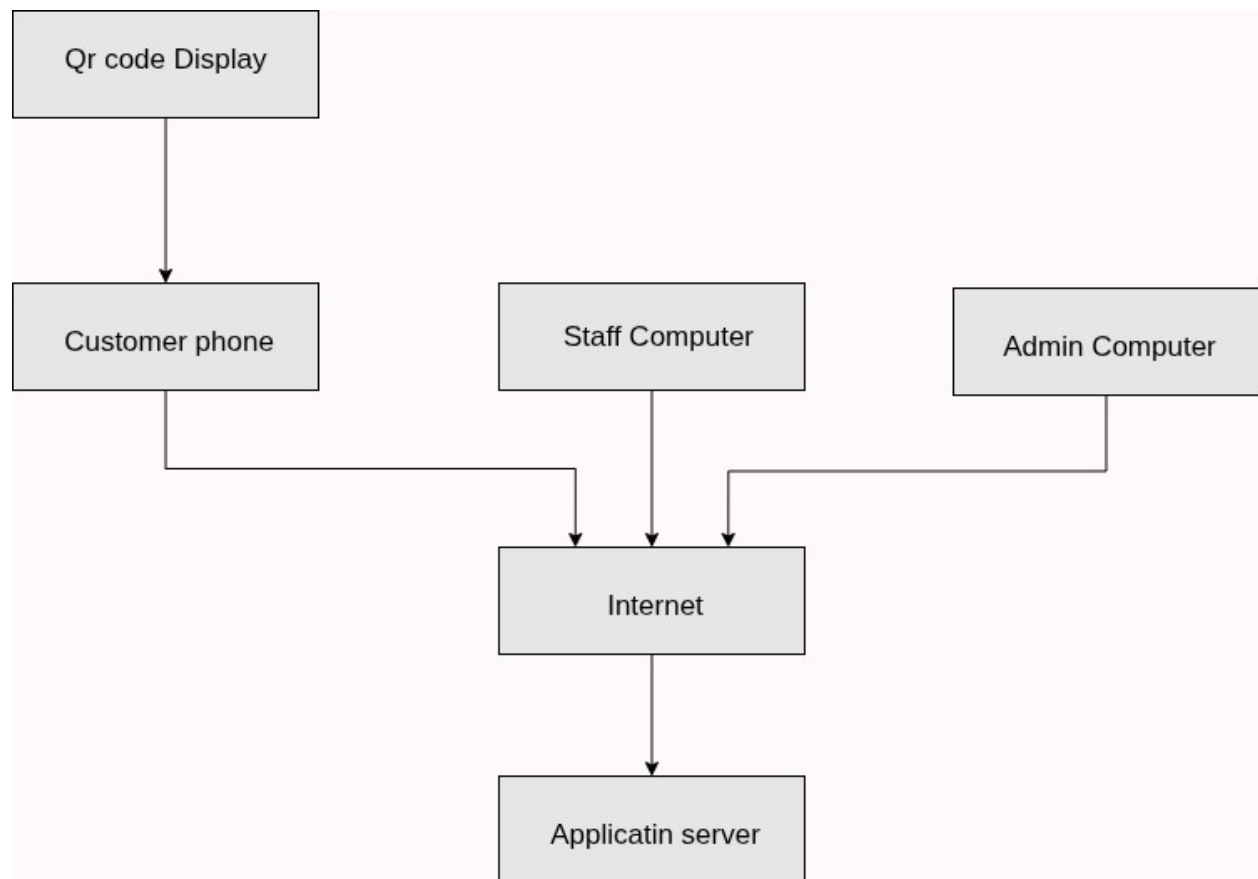


Figure 1 System Hardware Architecture

2.2 System Software Architecture

The system follows a three-tier software architecture consisting of frontend, backend, and database layers.

Presentation Layer: This is user interface which is play role for User interaction, display queue information and Send requests to backend. Technology used HTML, CSS, JavaScript React.js and React native.

Application Layer: This is layer which handle the business logic processing, Queue management, Authentication and authorization, API handling. Technology used is Django rest framework

Data Layer: This is where the system data stored, Manage data integrity and Support queries and reporting. Technology used is MySQL

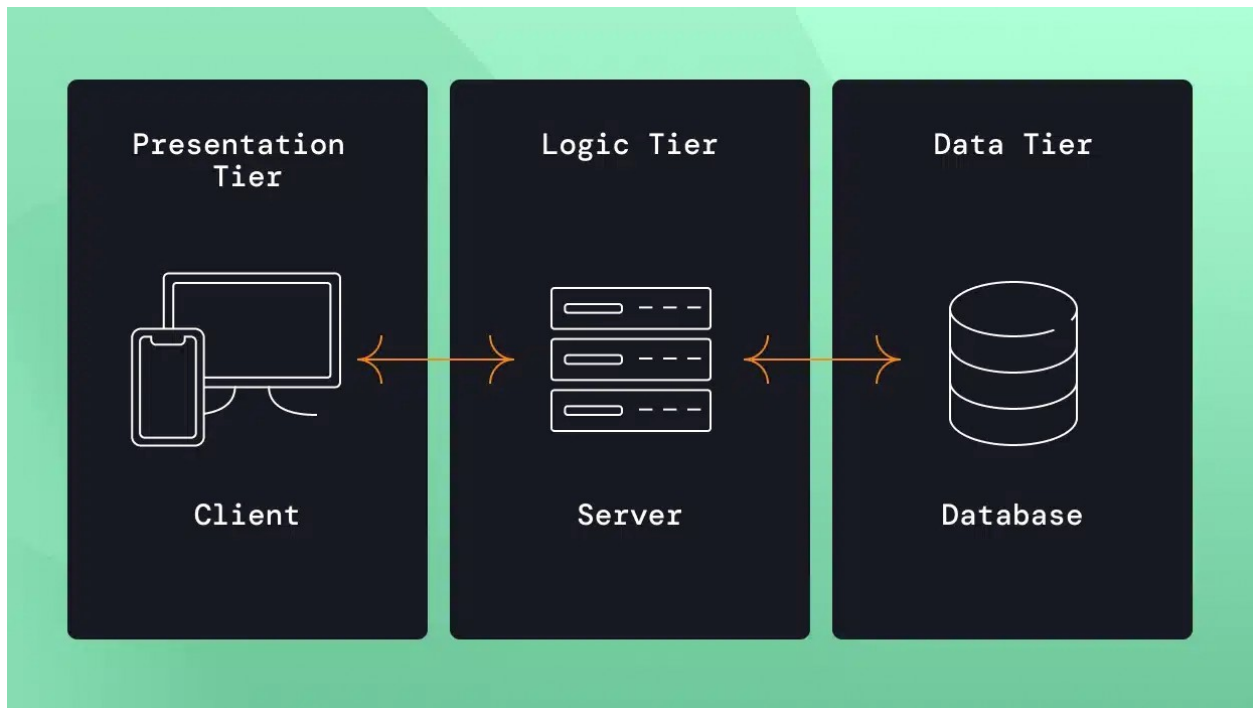


Figure 2 System Software Architecture

2.3 Internal Communications Architecture

Communication Between Frontend and Backend: The system uses RESTful APIs, Communication via HTTP/HTTPS protocols and all data is exchanged in JSON format

Communication Between Backend and Database: Backend communicates with database using Django ORM where all SQL queries are executed internally

User Interaction Flow

1. User scans QR
2. Frontend sends request to backend
3. Backend processes request
4. Backend interacts with database
5. Response is sent back to frontend
6. Frontend displays result to user

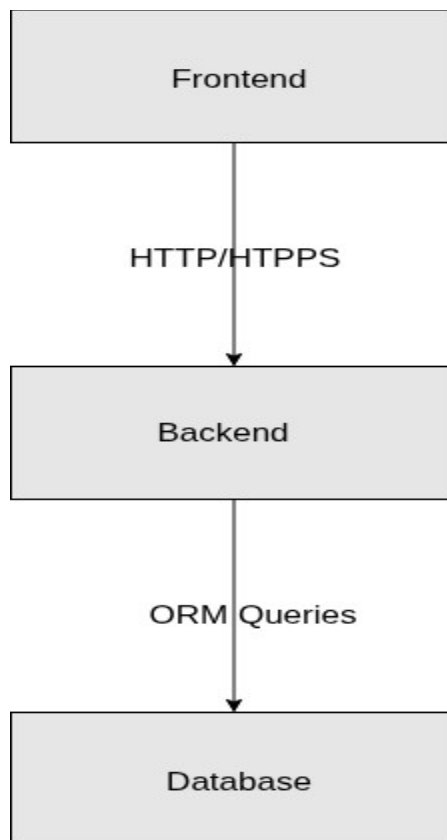


Figure 3 Internal Communication Architecture

FILE AND DATABASE DESIGN

3.1 Database Management System Files

The QueueXpress system uses MySQL Database Management System (DBMS) to store structured data.

The database is designed using relational principles and consists of the following main tables:

User Table

Stores all system users including staff and administrators.

FIELD	TYPE	DESCRIPTION	CONSTRAINT
id	INT (PK)	Unique identifier	PRIMARY KEY, AUTO_INCREMENT
name	VARCHAR	Full name	NOT NULL
phone	VARCHAR	Contact number	NOT NULL
role	ENUM (Staff or Admin)	User role for permission	NOT NULL, DEFAULT 'staff'
password	VARCHAR	Encrypted password	NOT NUL
created at	DATETIME	Account creation date	DEFAULT CURRENT_TIMESTAMP

Service Table

Stores available services offered by the organization.

Field	Type	Description	CONSTRAINT
id	INT (PK)	Service ID	PRIMARY KEY, AUTO_INCREMENT
name	VARCHAR	Service name	NOT NULL
description	TEXT	Service description	

Queue Table

Stores all queue entries generated when customers join the queue.

Field	Type	Description	CONSTRAINT
id	INT (PK)	Unique queue record ID	PRIMARY KEY, AUTO_INCREMENT
queue number	INT	Sequential queue number	NOT NULL
batch number	INT (FR)	Batch group number	NOT NULL
served	INT (FR)	Selected service	NOT NULL
phone_number	VARCHAR(15)	Customer phone number	NOT NULL
status	ENUM	waiting / active / served / skipped	NOT NULL, DEFAULT 'waiting'
created at	DATETIME	Time customer joined queue	DEFAULT CURRENT_TIMESTAMP
updated at	DATETIME	Last status update time	NULL
called_at	DATETIME	Time when customer was called by staff	NULL
served_at	DATETIME	Time when customer was served	NULL
served_by	INT (FK)	Staff who served the customer	NULL

Batch Table

Stores batch group information used for queue control.

Field	Type	Description	CONSTRAINT
id	INT (PK)	Batch ID	PRIMARY KEY, AUTO_INCREMENT
batch number	INT	Batch identifier	NOT NULL
Max size	INT	Number of users per batch	NOT NULL
status	ENUM	pending / active / completed	NOT NULL, DEFAULT 'pending'
batch_limit	INT	Max customers per batch	NOT NULL

Setting Table

Stores system settings for queue control.

Field	Type	Description	CONSTRAINT
id	INT (PK)	Batch ID	PRIMARY KEY, AUTO_INCREMENT
batch size	INT	Number of users per batch	NOT NULL
Queue reset	datetime	Time to reset queue	NOT NULL
created at	DATETIME	Time updated	DEFAULT CURRENT_TIMESTAMP
updated_at	DATETIME	Last update time	DEFAULT CURRENT_TIMESTAMP

Feedback Table

Stores customer feedback after service.

Field	Type	Description	CONSTRAINT
id	INT (PK)	Feedback ID	PRIMARY KEY, AUTO_INCREMENT
Queue id	INT (FK)	Linked queue entry	NOT NULL
message	TEXT	Feedback message	
rating	INT	Rating (1–5)	
created at	DATETIME	Time submitted	DEFAULT CURRENT_TIMESTAMP

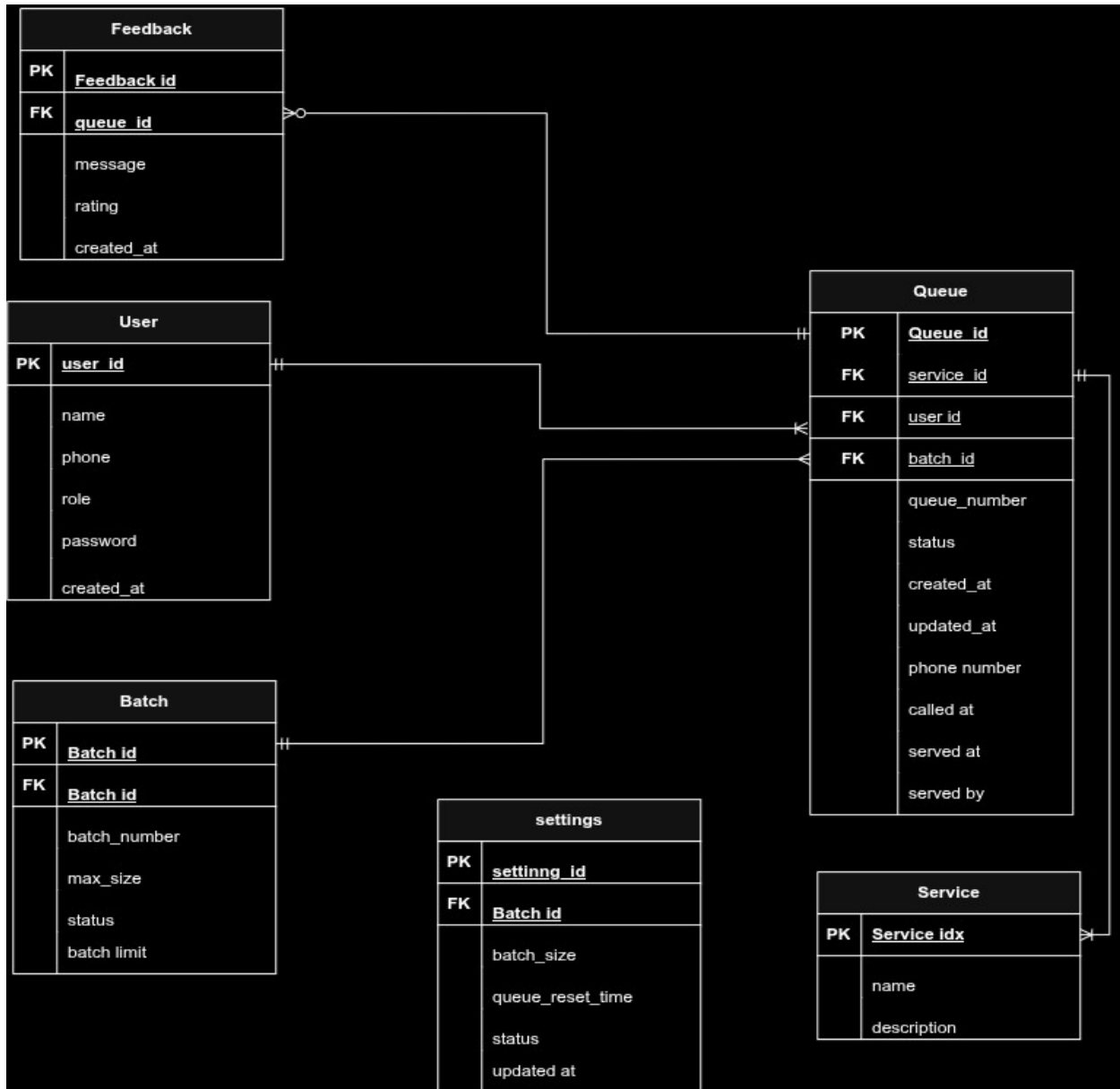


Figure 4 Entity Relationship Diagram

3.2 Non-Database Management System Files

QueueXpress uses the following non-DBMS files:

- i. Configuration Files (.env) which - Stores sensitive system settings such as Database credentials, Secret keys and Environment variables
- ii. Static Files Used for user interface design such as CSS stylesheets, JavaScript files and Images and icons
- iii. QR Code Assets Used for queue entry

- iv. Backup Files Used for recovery in case of failure such as Database backup files

HUMAN-MACHINE INTERFACE

4.1 Inputs

Customer Inputs: Customers interact with the system without registration using simple inputs.

QR Code Scan Used to join the queue by scanning a QR code, Feedback Submission which include Text message input and Rating selection

Staff Inputs: Staff interact with the system through a dashboard interface. Call Next Customer Button to move queue forward, Serve Customer which Marks current customer as served and Skip Customer which Skips absent customer

Administrator Inputs: Administrators configure and manage the system. User Management Inputs Which Add staff/admin, Edit user details. Also, System Configuration Inputs include Set batch size Update system settings

Input Validation

- All inputs are validated to ensure correctness:
- Required fields must not be empty
- Rating must be within allowed range (1–5)
- Invalid QR codes are rejected

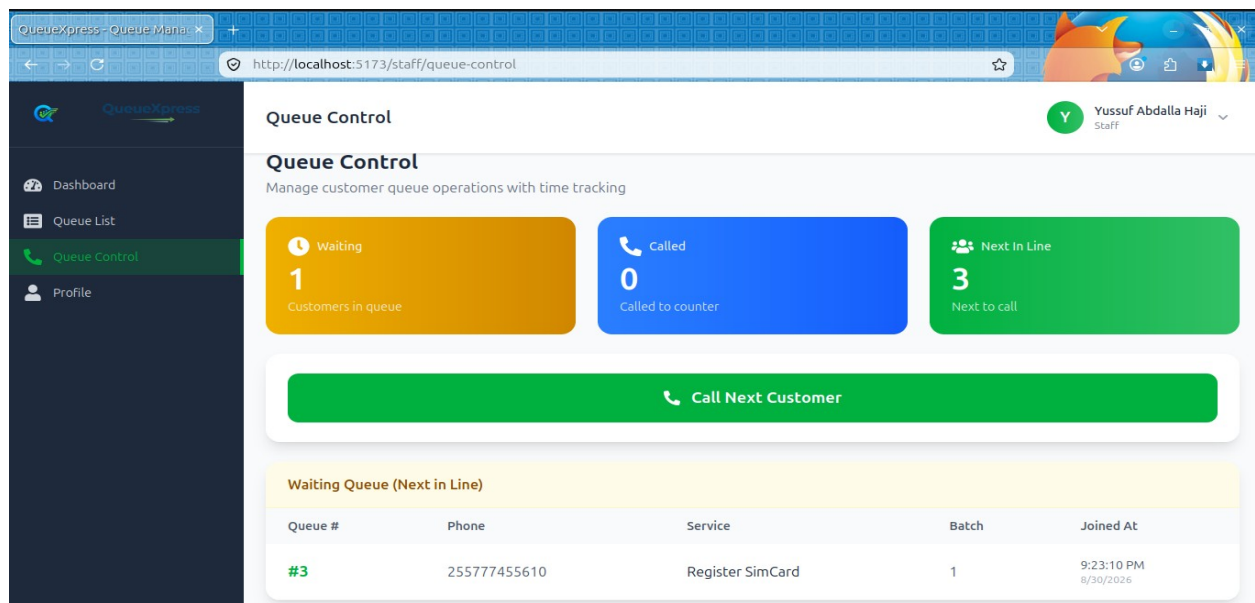
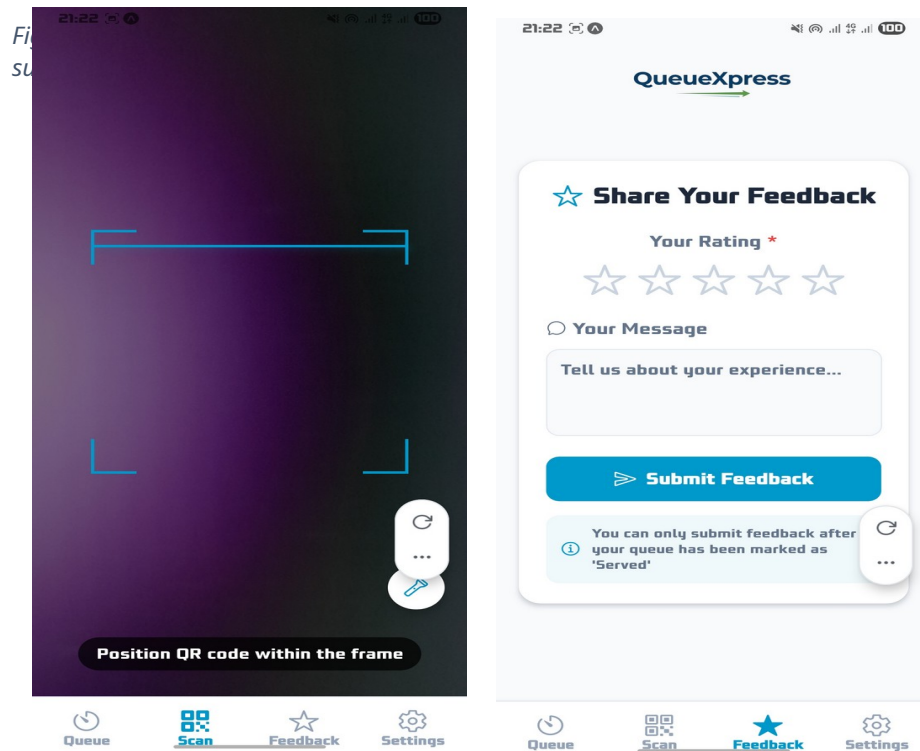


Figure 6 staff dashboard

Figure 7 admin configure page

4.2 Outputs

QueueXpress generates outputs for customers, staff, and administrators

Customer Outputs: After joining the queue, customers receive Queue Number, Batch Number, estimated waiting time, Queue Status (waiting, active, served, skipped) and People Ahead

Staff Outputs: Staff dashboard displays Current queue number, List of upcoming customers, Queue progress

Administrator Outputs: Admin interface provides System usage reports, Number of customers served and Performance statistics, Average Response Time (time from called to served)

System Messages: The system provides feedback messages such as Successful queue entry and Error messages

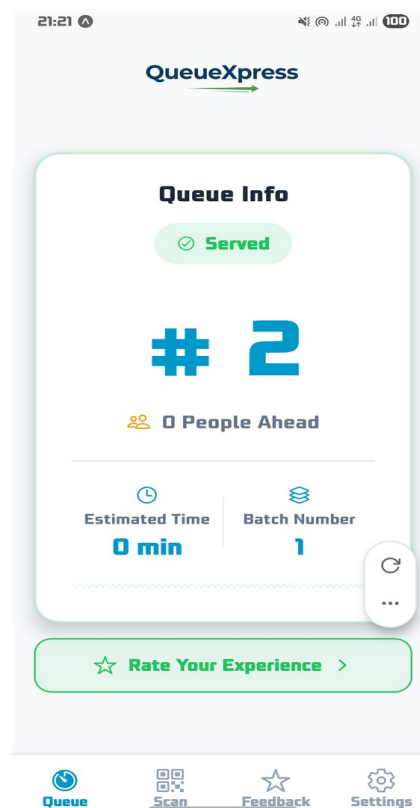


Figure 8 queue details

DETAILED DESIGN

5.1 Hardware Detailed Design

Client Devices

These devices are used by end users to access the system.

a) Customer Devices

- Smartphones with camera capability
- Used for scanning QR codes and viewing queue status

b) Staff Devices

- Desktop or laptop computers
- Used to manage queue operations through dashboard

c) Administrator Devices

- Desktop or laptop computers
- Used for system configuration and monitoring

QR Code Infrastructure

- Printed QR code or digital display
- Positioned at service location
- Used as entry point to the system

Network Requirements

- Internet connectivity required

5.2 Software Detailed Design

5.2.1 System Modules

1. Queue Entry Module

This module handles the core operation of adding a customer to the queue. When a customer scans the QR code, this module is invoked. It retrieves today's highest queue

number, increments it, calculates the batch number, creates a new queue record, and returns the assigned numbers to the customer.

2. Batch Management Module

This module handles Controls grouping of customers. The batch is configured by admin with fixed number of user and when user scan Qr code automatically system assign it into current batch

3. Queue Control Module

This is used by staff. Staff member can click the "Call Next" button to call for the next customer and it updates that record's status to 'active', records the called timestamp. Also, staff can Mark a customer as served. This is typically called after the customer has completed their transaction. Updates status to 'served'. And finally, can click skip if the customer is absent in area

4. Feedback Module

Allows customers to submit feedback. Where the user must input the rate for services and feedback message. This will be stored anonymously using queue id to protect customer

5. User Management Module

This module is Managed by admin to control staff accounts include create, update, read or delete the user

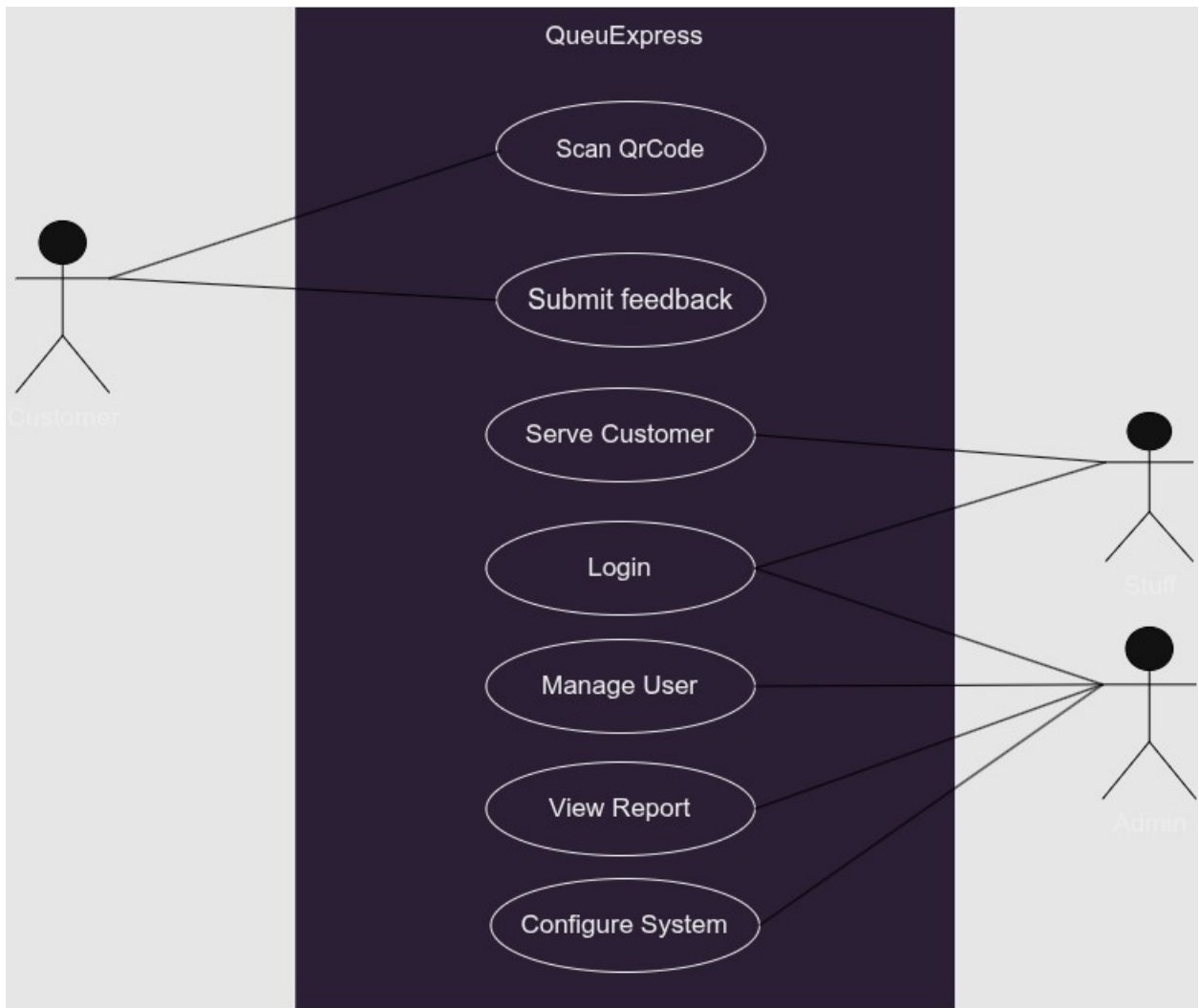


Figure 9 use case diagram

5.2.2 Processing Logic

a) Queue Assignment logic

1. Receive request from QR scan
2. Get last queue number
3. Increment queue number
4. Assign batch based on size
5. Check for duplicate active queue (same phone number with status 'waiting' or 'called')
6. Save record

7. Return queue details

b) Batch Allocation Algorithm

1. Count users in current batch
2. If count $< 10 \rightarrow$ assign same batch
3. Else $\rightarrow$ create new batch
4. Assign new batch number

c) Queue Processing Algorithm

1. Staff clicks "Call Next"
2. Select next waiting queue
3. Update status to "called"
4. Record called_at timestamp

d) Skip Logic

1. If customer absent
2. Staff clicks skip
3. Update status to "skipped"
4. Move to next queue

d) Serve Logic

1. Staff clicks "Serve"
2. Update status to "served"
3. Record served_at timestamp
4. Record served_by (staff ID)

5.3 Internal Communications Detailed Design

Frontend to Backend Communication

- Protocol: HTTP/HTTPS
- Format: JSON
- Method: REST API

Backend to Database Communication

- Uses Django ORM
- Executes SQL queries internally

Communication Flow

1. User action (scan or click)
2. Frontend sends API request
3. Backend processes logic
4. Backend queries database
5. Response sent to frontend
6. Frontend updates UI



Figure 10 Communication Diagram

Error Communication Handling

Backend returns error messages

Frontend displays user-friendly alerts

Security in Communication

HTTPS encryption

Input validation

Authentication for staff/admin

EXTERNAL INTERFACES

6.1 Interface Architecture

The system uses a client-server interface architecture, where users interact through the frontend, which communicates with the backend services.

User Interfaces

The system provides interfaces for:

Customers: Access interfaces via smartphone which enable to Scan QR code, join queue and View queue status

Staff: Staff Access Dashboard interface which enable to Manage queue operations

Administrators: Admin has Control panel interface which enable admin to Manage users and system settings

Hardware Interfaces

The system interacts with:

- Smartphone cameras for QR scanning
- Computers for staff and admin dashboards
- QR code display wither printed or digital

Software Interfaces

The system integrates internal software components:

- Frontend (React)
- Backend (Django)
- Database (MySQL)

Communication Interfaces

Communication is handled using RESTful APIs in JSON data format through HTTP/HTTPS protocols

6.2 Interface Detailed Design

Customer Interface Design

Input:

QR code scanning

Feedback submission

Output

Queue number

Batch number

Queue status

Notifications

Staff Interface Design

Input:

Call next

Serve customer

Skip customer

Output

Current queue

Next queue list

Queue updates

Administrator Interface Design

Input

- Add/edit users

- Configure system

Output:

Reports

System status

User management data

API Interface Design

The system uses REST APIs for communication.

Endpoint	Method	Description
/api/join/	POST	Customer joins queue
/api/queue/status/<queue_id>/	GET	Get queue status
/api/feedback/	POST	Submit feedback
/api/staff/call-next/	POST	Staff calls next customer
/api/staff/serve/<queue_id>/	POST	Staff serves customer
/api/staff/skip/<queue_id>/	POST	Staff skips customer
/api/staff/queue-list/	GET	Staff views waiting queues
/api/admin/dashboard-stats/	GET	Admin dashboard stats (includes response time)
/api/admin/report/	GET	Admin report data
/api/admin/feedback/	GET	Admin views feedback
/api/admin/settings/	PUT	Admin updates settings
/api/admin/services/	GET/ POST	Admin manages services
/api/admin/staff/	GET/ POST	Admin manages staff

Data Format

All data exchanged uses JSON format

Error Handling in Interfaces

Invalid requests return error messages

Network issues handled with retry mechanisms

User-friendly messages displayed

Security in Interfaces

- HTTPS communication
- Input validation
- Authentication for staff/admin

SYSTEM INTEGRITY CONTROLS

7.1 Internal Security

QueueXpress implements multi-layered security controls to protect sensitive information.

Authentication Controls:

Password Hashing: Passwords are hashed using bcrypt algorithm with

JWT Authentication: Access tokens expire after 24 hours, refresh tokens after 7 days

Token Management: Tokens stored in localStorage (web) and AsyncStorage (mobile)

Authentication: Admin uses username + password; Staff uses work_id + password

Session Fixation Prevention: Session identifier is regenerated after successful login

Login Rate Limiting: Planned for future implementation

HTTPS Enforcement

Authorization (Role-Based Access Control)

Customer will have full access into customer page only and restriction to the staff and admin dashboard

staff will have full access into customer and staff dashboard page and restriction to the admin dashboard

adman will have full access the whole system

7.2 Audit Procedures

The following events are logged in the logs table:

Authentication audit Login success, login failure, logout, password change

Queue Operations auditing Call next, serve, skip

User Management audit Staff creation, staff edit, staff disable, role change

System Configuration audit Batch size change, reset time change

7.3 Data Validation Controls

Client-side (JavaScript) Type checking, format validation, required field checking and Immediate feedback to user before submission

Server-side (Django Forms and Serializers) Type checking, format validation, length checking, range checking, business logic validation for Primary defense against invalid data |

Duplicate Queue Prevention: System checks if a phone number already has an active queue (status = 'waiting' or 'called') before creating a new entry. If a duplicate is found, the existing queue info is returned instead of creating a new one.

Database Column data types, constraints, foreign key integrity